V93000 Specifications

Link Scale PCIe4 card

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Notices

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Link Scale PCIe specifications

Use the Link Scale PCIe4 card to execute software-based functional tests and high-speed input/output scan tests on Smart Scale™ and EXA Scale™ test systems.

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Scope of specifications

Note	The specifications are subject to change without notice. Make sure to refer always to the latest revision found in the most recent SmarTest documentation (System Reference > Specifications).

This specification covers the Link Scale PCIe4 card (part number E8018PCI for Smart Scale™, EK806PCI for EXA Scale™).

Advantest specifies and verifies the specifications at the DUT interface pogo level. Appropriate DUT board design and layout including proper usage of the GND sense inputs and fixture delay measurements (TDR) is required to ensure specified performance at the device under test.

Specifications define the warranted technical performance covered by products within the stated adjustment interval and operating conditions. Typical specifications define the technical performance met by at least 80 % of all products during product qualification. A typical specification is not covered by product warranty and does not include measurement uncertainties.

Link Scale PCIe4 card overview

Link Scale PCIe4 is a new type of digital card for the V93000 Smart Scale™ and EXA Scale™ generation of SoC test systems. It supports the following use cases as part of engineering or production test programs:

- Download, execution and tracing of software-based functional test content in binary (executable) form
- Scan test with data delivery and result upload through a high-speed input/output (HSIO) PCI Express interface
- Basic DC test (continuity, impedance) through attached external resources.

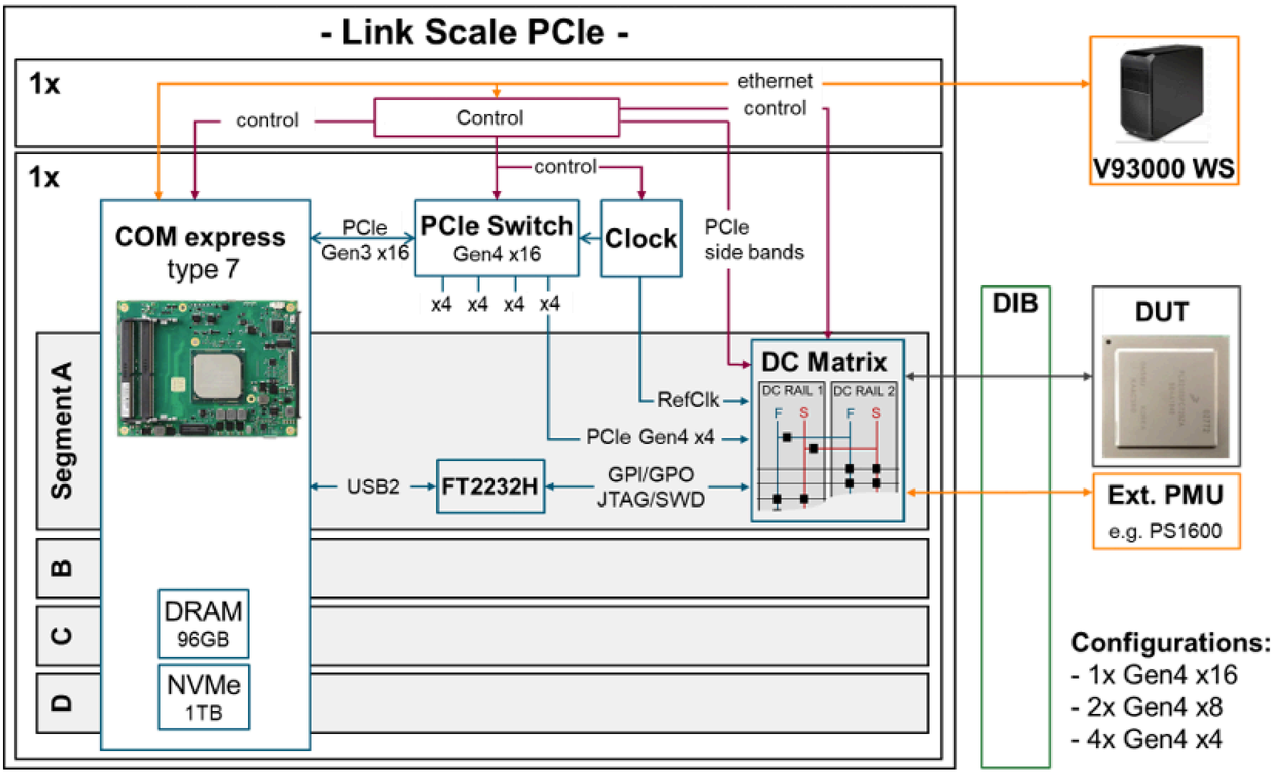
Link Scale PCIe4 card

The Link Scale PCIe4 card uses a computer-on-module (COM) architecture to provide industry standard interfaces (PCIe4, JTAG, SWD) with full protocol stack support. Each COM runs an operating system with support for user-specific drivers and executables. The COM interfaces with the system controller (SmarTest workstation) through a LAN connection and, if needed, trigger signals to/from other channels in the test system.

The COM connects to the DUT(s) through a PCIe switch that allows the card to support multiple PCIe links to be operated in parallel. Depending on the configuration, either one link with 16 lanes, two links with 8 lanes each, or four links with 4 lanes each are active. Using one link per site, up to four sites can tested in parallel.

The card also offers four AUX ports which can be used as low-speed protocol interfaces (JTAG / SPI / SWD) and as GPIOs to connect external DC resources (for example Pin Scale 1600 PPMU). A DC switch matrix on the card routes the external DC resources to the target signals.

The figure below shows a block diagram of the Link Scale PCIe4 card.



Block diagram of Link Scale PCIe4 card

Further information

For more information about the Advantest V93000 SoC Series, please visit *www.advantest.com*.

Contact information

For more information about the Advantest V93000 SoC cards, please contact your local Advantest sales representative.

This information is subject to change without notice.

Link Scale PCIe port specifications

Ports

Total number of channels	96
PCI Express Gen4 lanes	16 (16 differential Rx pairs and 16 differential Tx pairs), configurable as 1x16, 2x8, or 4x4
PCI Express side band signals	12 (4 x differential reference clock + 4 x reset)
AUX ports (JTAG / SPI / SWD / GPIO)	4 (with 5 signals per port)

Speed

PCI Express Gen4 lanes	16.0 Gbit/s (Gen4 mode)[1] 8.0 Gbit/s (Gen3 mode) 5.0 Gbit/s (Gen2 mode) 2.5 Gbit/s (Gen1 mode)
PCI Express side band signals	PCIe Gen4 compliant
AUX ports (JTAG / SPI / SWD / GPIO)	15 MBit/s

Levels

PCI Express Gen4 lanes	PCIe Gen4 compliant
PCI Express side band signals	PCIe Gen4 compliant [2]
AUX ports (JTAG / SPI / SWD / GPIO)	1.2 V, 1.5 V, 1.8 V, 2.5 V, 3.3 V [3]

[1] *Maximum PCIe Gen4 throughput can only be achieved if max. N=8 lanes are used simultaneously in Gen4 mode. If more than N=8 lanes are used in Gen4 mode, maximum throughput across all lanes will be equivalent to a PCIe Gen3x16 link.*

[2] *The PERST signal uses the same voltage as the AUX port. AUX voltage levels below 2.0 V may need a level shifter on the DUT board.*

[3] *The selected level applies to all signals of the AUX port as well as the PCIe reset signal (PERST).*

Link Scale PCIe environment and maintenance

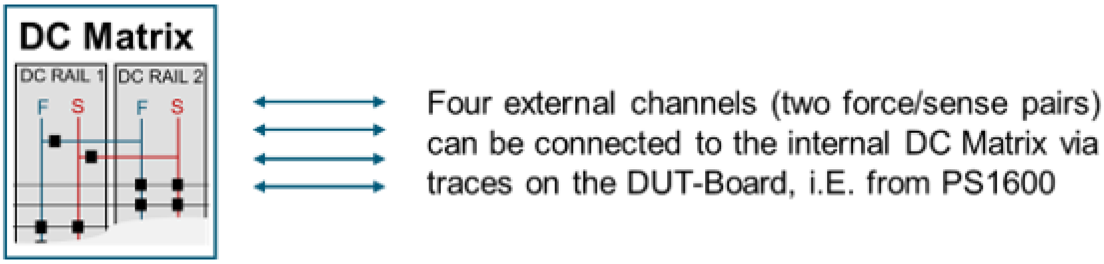
For the definition of week, month, year in the context of a maintenance period refer to *Units of maintenance periods*.

Environment, operation, and maintenance

Operating temperature range	+15 # to +30 # (+59 # to +86 #)
Temperature range for warranted specifications	+20 # to +30 # (+68 # to +86 #)
Maximum humidity at +30 # (+86 #)	< 70% relative humidity, non-condensing
Storage temperature range, without the presence of water	-40 # to +70 # (-40 # to +158 #)

Link Scale PCIe DC characteristics

The Link Scale PCIe card does not contain built-in resources for DC measurements. Instead, it uses an internal switch matrix between the AUX/GPIO signals and the other ports to connect external resources (for example PS1600 channels) for DC measurements such as continuity or (differential) impedance. DC access can be done single-ended or in a higher accuracy force/ sense configuration (two channels)



Link Scale PCIe4 DC matrix

Two DC rails (force/sense pairs) are accessible per site via the GPIO signals and can be used as follows:

- Two parallel one-terminal measurements per site, that is continuity
- Two parallel two-terminal measurements per site, that is leakage, with sensing
- One four-terminal measurement per site, that is differential impedance, with sensing

Accordingly, up to four external channels (or two force/sense pairs) are needed for DC measurements.

The signal path through the Link Scale card can reduce the accuracy of the DC measurements by the following terms maximum.

Voltage measure	10 mV
Current force / voltage measure	10 mV + iForce * 25 Ω
Leakage current	20 nA

Link Scale PCIe module PC (COM) specifications

Module PC	
CPU	Intel Xeon D1548 eight-core
DRAM	96 GByte (3 x 32 GByte Dual Channel DDR4-2400)
Mass Storage	950 GByte NVME